AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT141
Introduction to Probability and Statistics

Final Examination
Spring Semester 2025/2026

Date: 3 / 6 /2026

Number of Exam Pages: (4) (including this cover sheet)	**Time Allowed: 2 Hours**

Instructions:

Please read the following instructions before starting:

- Total Number of Questions:5
- This is a closed book exam.
- Be sure you write your name and ID on the External Answer booklet.
- This exam accounts to 50% of the total mark in the course.
- Budget your time according to the mark assigned to each question.
- Mobile phones and all other mobile communication equipment are NOT allowed.
- Calculators are allowed.

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Part 1: MULTIPLE CHOICE QUESTIONS

Solve all the questions. Each question is worth 2 marks, giving a total of 10 possible marks.

Q–1: A student scored 75 in an exam where the mean was 60 and the standard deviation was 5. What is the Z-score?

a) 4
b) 5
c) 2
d) 3

Q–2: Which of the following is a property of the normal distribution?

a) It is skewed to the right
b) The mean and median are different
c) It is symmetric around the mean
d) The total area under the curve is 2

Q–3: A factory produces light bulbs, the probability of a defective bulb is 0.10. If 50 bulbs are tested, what is the variance?

a) 4.5
b) 5
c) 45
d) 1.2

Q–4: A class of 30 students has a mean score of 70. A new student joins with a score of 100. What is the new mean?

a) 72
b) 70
c) 71.2
d) 70.97

Q–5: A dataset has 5 values. The mean is 20 and 4 of the values are: 10, 15, 25, 30 What is the 5th value?

a) 18
b) 22
c) 20
d) 25

Part 2: ESSAY QUESTIONS

Each question is worth 10 marks. You should answer all FOUR questions, giving a total of 40 possible marks.

Q–1: [3+2+5 Marks]

An airline knows that **5% of passengers** who book a seat do not show up for their flight. A gate agent randomly selects **6 booked passengers** from a flight manifest. The agent wants to determine the probability that:

I. Exactly **3 passengers** do not show up.

II. **No passengers** fail to show up.

III. **At least 2 passengers** do not show up.

Q-2: [3+3+4 Marks]

A continuous random variable with probability density function $f(x)$ **given by**

$f(x) = k\ x^2 \qquad 0 \leq x \leq 2$ **where k is a positive constant.**

a) **Calculate the value of k.**
b) **Find P(X < 1).**
c) **Find P(1< X < 1.5).**

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Q–3: [5+5 Marks]

a) **A academic performance analyst at a university is analyzing the** weekly study hours of students**, which follows a** normal distribution **with a mean of** 1,000 minutes **and a standard deviation of** 50 minutes**. She wants to know what percentage of students are expected to study between** 950 minutes and 1,050 minutes **per week.**

Hint: Use the table below

z	-1	-2.00	1.17	1.18	1	2.00	1.80	3.00
P (Z < z)	0.1587	0.0228	0.8790	0.8810	0.8413	0.9772	0.9641	0.9987

b) **A leading telecom company is reviewing the performance of its newly developed smartphone batteries. A sample of** 25 batteries **showed an average lifespan of** 1,200 hours**. Assuming the population standard deviation is known to be** 80 hours**, construct a** 95% confidence interval **to estimate the true mean battery lifespan.**

[Given: $Z_{0.025} = 1.96$]

Q–4: [10 Marks]

A hospital quality control officer is investigating whether the average systolic blood pressure **of patients treated at a clinic has changed from the established baseline value of** 500 mmHg ***(scaled units for consistency)*****. She selects a random sample of** 40 patients **and finds the sample mean to be** 495 units**. Given that the population standard deviation is** 10 units**, test the hypothesis at the** 0.05 significance level **to determine if there is evidence of a difference in the average blood pressure reading.**

[Given: $Z_{0.025} = 1.96$]

End of questions

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